reality are forever beyond our ability to know simultaneously, even in principle? Or does quantum uncertainty break the classical mold completely, telling us that the list of attributes our classical intuition ascribes to reality, a list headed by the positions and velocities of the ingredients making up the world, is misguided? Does quantum uncertainty tell us that, at any given moment, particles simply do not possess a definite position and a definite velocity?

To Bohr, this issue was on par with a Zen koan. Physics addresses only things we can measure. From the standpoint of physics, that *is* reality. Trying to use physics to analyze a "deeper" reality, one beyond what we can know through measurement, is like asking physics to analyze the sound of one hand clapping. But in 1935, Einstein together with two colleagues, Boris Podolsky and Nathan Rosen, raised this issue in such a forceful and clever way that what had begun as one hand clapping reverberated over fifty years into a thunderclap that heralded a far greater assault on our understanding of reality than even Einstein ever envisioned.

The intent of the Einstein-Podolsky-Rosen paper was to show that quantum mechanics, while undeniably successful at making predictions and explaining data, could not be the final word regarding the physics of the microcosmos. Their strategy was simple, and was based on the issues just raised: they wanted to show that every particle does possess a definite position and a definite velocity at any given instant of time, and thus they wanted to conclude that the uncertainty principle reveals a fundamental limitation of the quantum mechanical approach. If every particle has a position and a velocity, but quantum mechanics cannot deal with these features of reality, then quantum mechanics provides only a partial description of the universe. Quantum mechanics, they intended to show, was therefore an incomplete theory of physical reality and, perhaps, merely a stepping-stone toward a deeper framework waiting to be discovered. In actuality, as we will see, they laid the groundwork for demonstrating something even more dramatic: the nonlocality of the quantum world.

Einstein, Podolsky, and Rosen (EPR) were partly inspired by Heisenberg's rough explanation of the uncertainty principle: when you measure where something is you necessarily disturb it, thereby